

ProtoDUNE-HD Wire-Cell 3-D Imaging & Deghosting · per APA

tomographic tiling: fired U/V/W wire strips coincide → blobs (+ false crossings) → charge-solve & deghost → 3-D clusters

SP frames
per APA
gauss / wiener

① Slice

MaskSlices — time-slice
per plane, threshold charge
→ fired U/V/W activity / slice
multi-pass: UVW · UV · VW · UW

② Tile — RayGrid

GridTiling, per face 0/1
U/V/W strips' triple overlap
→ 2-D blob per slice
false crossings → ghost blobs

③ Solve & Deghost

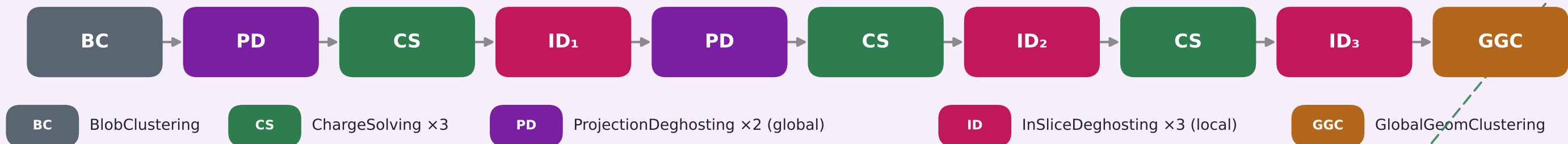
BlobClustering: stack blobs
across slices → 3-D clusters
then charge-solve ≠ deghost
ladder (3 rounds) ▼ below

④ ClusterFileSink
→ clusters-apa-
-ms-active.tar.gz
(3-view live blobs,
post-deghost)

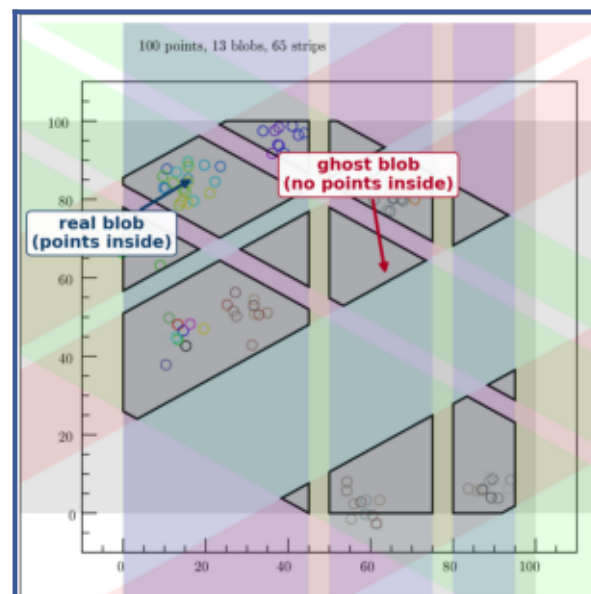
pre-proc: CMMModifier ·
FrameMasking · ChargeError

⊕ dead / masked fork: 2-view blobs (dummy plane, span 1500), geometry only → clusters-apa-*-ms-masked.tar.gz

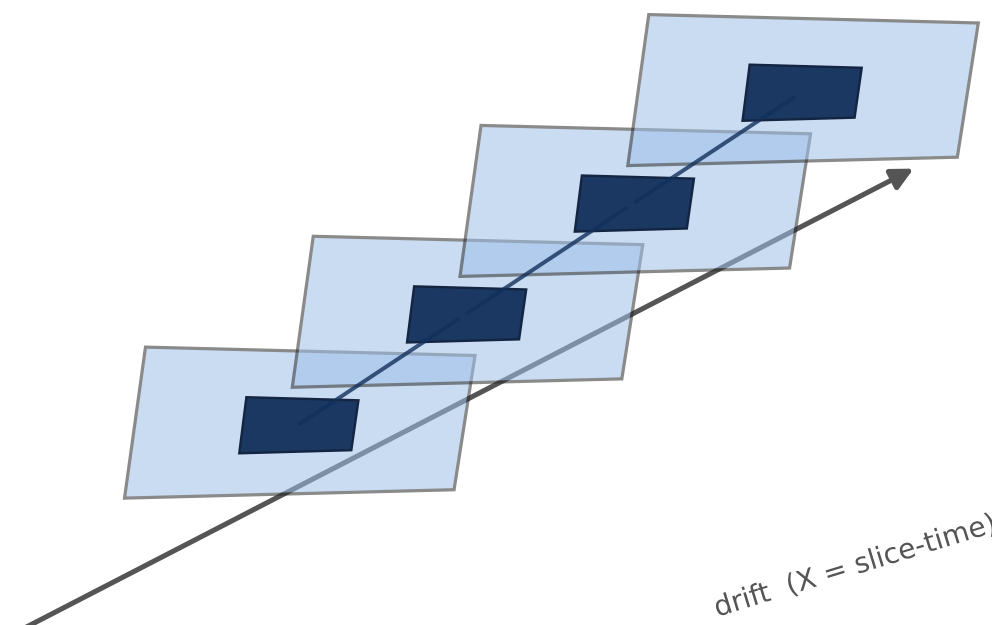
inside ③ · charge-solve ≠ deghost ladder (uboone-solving) — note the asymmetry: ProjectionDeghosting ×2, ChargeSolving ×3, InSliceDeghosting ×3



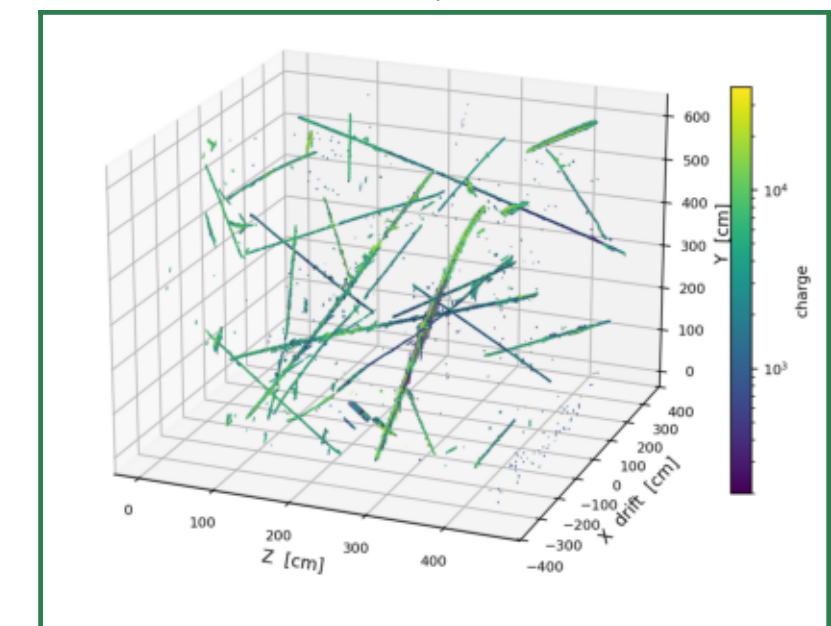
② tiling: false triple-overlaps ⇒ ghosts



ghost concept — toy 3-plane tiling (Fig. 8)



stack 2-D blobs along drift → 3-D image



full event 3-D charge display (Bee) — deghosted result